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6.(a)

令 $x^2=\sum_{l=0}^{\infty} c_l P_l(x)$

$\Rightarrow \quad c_l=\frac{2l+1}{2}\int_{-1}^{1} x^2 P_l(x)\,dx$

$\Rightarrow l=0$ 时, $c_1=\frac{1}{2}\int_{-1}^{1} x^2 P_0(x)\,dx=\frac{1}{3}$

$l=2$ 时, $c_2=\frac{5}{2}\int_{-1}^{1} x^2\cdot P_2(x)\,dx=\frac{25}{24}$

$\Rightarrow \quad x^2=\frac{1}{3}P_1(x)+\frac{25}{24}P_2(x)$

(b) 由 5(b),

$$\int_0^1 P_k(x)P_l(x)\,dx=\begin{cases}\dfrac{[P_k'(x)P_l(x)-P_l'(x)P_k(x)]}{k(k+1)-l(l+1)}, & k\neq l\\ \dfrac{1}{2l+1}, & k=l\end{cases}$$

$\Rightarrow$ 令 $f(x)=\sum_{l=0}^{\infty} c_l P_l(x)$

$\Rightarrow \quad \int_{-1}^{1} f(x)P_k(x)\,dx=\sum_{l=0}^{\infty}\int_{-1}^{1} c_l P_l(x)P_k(x)\,dx=\frac{2}{2l+1}c_l$

$\Rightarrow \quad c_l=\frac{2l+1}{2}\int_0^1 x P_l(x)\,dx=\frac{2l+1}{2}\int_0^1 P_1(x)P_l(x)\,dx$

$$\Rightarrow c_l=\begin{cases}1, & l=1\\ \dfrac{2l+1}{2}\dfrac{[P_l(x)-xP_l'(x)]}{2-l(l+1)}, & l\neq 1\end{cases}$$

$$2xP_{l-1}(x)=\frac{(P_l(x)+(l-1)P_{l-2}(x))}{2l-1}P_{l-1}(x)+P_l(x)$$

d)

$$f(x)=(1-2xt+t^2)(1-2xt+t^2)^{-\frac{1}{2}}=(1-2xt+t^2)\sum_{l=0}^{\infty}P_l(x)t^l$$

$$=\sum_{l=0}^{\infty}\left[P_l(x)t^l-2xP_l(x)t^{l+1}+P_l(x)t^{l+2}\right]=\sum_{l=0}^{\infty}\frac{1}{2l-1}\left[-lP_l(x)+lP_{l-2}(x)+(2l-1)P_{l-1}(x)\right]t^l$$

7. (a) ① $k<l$ 时,

$$\int_{-1}^{1} P_k'(x)P_l(x)dx = 0$$

② $k>l$ 时

$$\int_{-1}^{1} P_l(x)P_k'(x)dx = P_k(x)P_l(x)\Big|_{-1}^{1} - \int_{-1}^{1} P_k(x)P_l'(x)dx$$

$$= 1-(-1)^{k+l}$$

(b) $$\int_{-1}^{1} P_l(x)(1-2xt+t^2)^{-\frac{3}{2}}dx = \int_{-1}^{1} P_l(x)\frac{1}{t}\sum_{k=0}^{\infty} P_k'(x)t^k dx$$

$$= \sum_{k=l+1}^{\infty}\left[1-(-1)^{k+l}\right]t^{k-1} = 2\sum_{k=0}^{\infty} t^{l+2k}$$

9.

分离变量解得一般解.

$$u(r,\theta) = \sum_{l=0}^{\infty}\left[A_l r^l + B_l r^{-l-1}\right]P_l(\cos\theta)$$



$$\begin{cases} u|_{r=a} = u_0 \\ u|_{r=b} = u_0\cos\theta, \text{ 同上,} \end{cases}$$

$$\begin{cases} \left[A_l a^l + B_l a^{-l-1}\right]\frac{2}{2l+1} = \int_{\pi}^{0} u_0 P_l(\cos\theta)d(\cos\theta) = \begin{cases} 2u_0, & l=0 \\ 0, & l\geq 1 \end{cases} \\ \left[A_l b^l + B_l b^{-l-1}\right]\frac{2}{2l+1} = \int_{\pi}^{0} u_0\cos\theta P_l(\cos\theta)d(\cos\theta) = \begin{cases} \frac{2u_0}{2l+1}, & l=0,1 \\ 0, & l\geq 3 \end{cases} \end{cases}$$

得:

$$\begin{cases} (A_1 + B_1 a^{-1})\cdot 2 = 2u_0 \\ (A_1 + B_1 b^{-1})\cdot 2 = 2u_0 \end{cases} \qquad \begin{cases} a^2 A_2 + B_2 a^{-1} = 0 \\ b^2 A_2 + B_2 b^{-1} = \frac{2u_0}{5} \end{cases}$$

$$\Rightarrow A_1 = u_0,\ B_1 = 0,\ A_2 = -\frac{2u_0}{5}\frac{b^3}{a^5-b^5},\ B_2 = \frac{2u_0}{5}\frac{a^5b^3}{a^5-b^5}$$

$$\Rightarrow u = u_0 - \frac{u_0}{5}\frac{b^3}{a^5-b^5}\left[r^2 - a^5 r^{-3}\right](3\cos\theta - 1)$$